Question 1: For the following proofs, suppose F is a field, and $a, b \in F$.

(a) If $a = 0$ or $b = 0$, we are done. If not, WLOG, let $a \neq 0$. Then, multiplying by a^{-1} (which is guaranteed to exist by A13), we get $b = 0$. Thus, $a = 0$ or $b = 0$.

(b) Suppose $a^2 = b^2$. Then, $a^2 - b^2 = (a - b)(a + b) = 0$. Using the result from (a), $a - b = 0$ or $a + b = 0$. Thus, $a = b$ or $a = -b$.

Question 2: should be Q2

Proof. Consider the set L, which is the set containing all lower bounds. It is bounded above by every element in S (which exist because by assumption, S is non-empty). So, L has a supremum l. We claim that $l = \inf S$.

Lower Boundedness By definition, l is a lower bound.

Greatness By definition, l is the greater than every lower bound.

Thus, $l = \inf S$, and therefore S admits an infimum. □

Question 3: should be Q3

(a) *Proof.* □

(b) We claim that $b = \sup S$.

Upper Boundedness Rewriting S as $S = \{ x \in \mathbb{R} \mid a \leq x < b \}$, we can see that b is an upper bound by definition.

Leastness For the sake of contradiction, assume b is not the supremum. Then, there exists a $\sup S = u$. By definition, we must have $a \leq u < b$. By Theorem 6.12 in the notes, we can construct a rational number r with $u < r < b$. By extension, we have $a \leq u < r < b$. So, $r \in S$, and u is not an upper bound. This a contradiction with our intial assumption, so b must be the least upper bound.

Thus, $\sup S = b$.

Question 4: should be q4. We compute that $\sup S = 1$ and $\inf S = 0$. We first prove the former.

Proof. The first element in S is 1, and every succesive element is smaller, thus 1 is the max of S. So, $1 = \sup S$. □

Next, we prove $\inf S = 0$.

Proof. how tf u do ts i thought the rationals were incomplete □

Question 5: should be q5.

Proof. Suppose α is irrational, and $r \in \mathbb{Q} \setminus 0$. Suppose for the sake of contradiction that $r\alpha \in \mathbb{Q}$. Then, we can write $r\alpha = \frac{c}{d}$ for $c, d \in \mathbb{Z}$. We can also write $r = \frac{a}{b}$ for $a, b \in \mathbb{Z}$. Then, we have $r\alpha = \frac{a}{b}\alpha = \frac{c}{d}$. Rearranging, we get that

$$\alpha = \tfrac{bc}{ad},$$

which means $\alpha \in \mathbb{Q}$ (since $a, b, c, d \in \mathbb{Z}$). However, this contradicts the assumption that α was irrational. Thus, $r\alpha$ is irrational for every $r \in \mathbb{Q} \setminus 0$. $\square$

Question 6: should be q7

(a) We compute x_3 in base 10 with this expression:

$$\sum_{n=1}^{\infty} \tfrac{2}{3^{3n}} = \tfrac{2}{3} \sum_{n=1}^{\infty} \tfrac{1}{3^n} = 0.\overline{3}_{10}.$$

It is in the Cantor set.

(b) Similarly for y_3:

$$\sum_{n=1}^{\infty} \tfrac{1}{3^{2n}} + \sum_{n=1}^{\infty} \tfrac{1}{3^{3n}} = \tfrac{1}{9} \sum_{n=1}^{\infty} \tfrac{1}{3^n} + \sum_{n=1}^{\infty} \tfrac{1}{3^n} = \tfrac{4}{54} = 0.0\overline{740}_{10}.$$

you tell me if its in the cantor set twin

(c) For z_3, there is only one digit to compute. It is in the third place to the right, so it is $\frac{1}{3^3} = \frac{1}{27} = 0.0\overline{370}_{10}$